AI-Powered Organ Matching for the OPTN

Research Landscape, Technical Approach, and Implementation Strategy

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1. The Problem: Why Current Matching Falls Short

The current OPTN organ allocation system relies on a set of organ-specific scoring formulas — MELD/PELD for liver, KDPI/EPTS for kidney, Lung CAS for lung, status-based urgency for heart — that were designed as linear or semi-linear models using a small number of clinical variables. These models have known limitations.

They use a fraction of available data. MELD, for example, uses only three lab values (bilirubin, INR, creatinine) plus sodium. The OPTN database captures hundreds of donor and recipient variables — demographics, comorbidities, sensitization profiles, surgical history, lab panels, imaging, geographic data, center-level performance — but the current scoring models cannot integrate this richness. Important interaction effects between donor and recipient characteristics go unmodeled.

They predict the wrong thing. Most current models predict a single outcome — typically waitlist mortality or a crude severity score. But the optimal allocation decision depends on a much more complex question: *given this specific organ and this specific patient, what is the expected post-transplant survival benefit compared to the patient's expected survival if they continue waiting?* This is a donor-recipient pair-level prediction, not a patient-only prediction.

They cannot capture nonlinear relationships. The relationship between clinical variables and transplant outcomes is highly nonlinear — for example, the interaction between donor age, cold ischemia time, and recipient comorbidities creates complex risk surfaces that linear models cannot represent. ML models trained on historical OPTN/SRTR data consistently outperform traditional scoring systems in predictive accuracy.

They are slow to adapt. Updating MELD or KDPI requires years of committee deliberation, public comment, and policy change. The clinical landscape evolves faster: new preservation technologies (normothermic perfusion), changing donor demographics (older donors, DCD donors, HCV-positive donors), and evolving immunosuppression protocols all shift the outcome surface, but the scoring models lag behind.

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2. State of the Research

2.1 ML Models That Outperform Current Scoring

Multiple peer-reviewed studies have demonstrated ML models with superior predictive performance over current OPTN scoring systems.

OPOM (Optimized Prediction of Mortality) — Liver. Developed at MIT by Bertsimas et al. (2019), OPOM uses optimal classification tree models trained on SRTR data to predict 3-month waitlist mortality. In simulation using the Liver Simulated Allocation Model, OPOM-based allocation reduced annual waitlist mortality by approximately 418 deaths compared to MELD-based allocation — a roughly 18% reduction. OPOM showed improved survival across all candidate demographics and geographic regions, and delivered higher AUC values across all disease severity groups. It was independently validated on a Canadian cohort of over 2,300 patients, achieving an AUC of 0.81 for 3-month dropout prediction versus 0.77 for MELD and 0.78 for MELD-Na.

UK-DTOP (UK Deceased Donor Kidney Transplant Outcome Prediction). This AI-based tool for kidney transplant outcome prediction achieved a discriminative power (AUCROC) of 0.74 compared to 0.62 for the UK-KDRI, representing a substantial improvement. The model uses AI decision-based approaches that handle missing data without imputation, and identified a fifth donor risk cluster not captured by the traditional KDRI quartile system — suggesting donor characteristics that existing models miss.

Kidney Graft Survival Models. Analysis of over 50,000 US kidney transplants demonstrated that ML models using deep learning autoencoders for dimensionality reduction improved prediction of graft failure across 1-year, 5-year, and longer timeframes. Meta-analyses of ML models in kidney transplantation show that hybrid ML models, support vector machines, and random forest-based approaches consistently outperform traditional Cox regression and logistic regression for graft survival prediction. Neural network models combined with SHAP (SHapley Additive exPlanations) provide richer clinical interpretability than Cox proportional hazards models while matching or exceeding predictive performance.

Stanford DCD Liver Model. Stanford Medicine researchers developed an ML model that predicts donor death timing after withdrawal of life support for donation after circulatory death (DCD) cases. The model reduced the rate of futile transplant procurements by approximately 60%, directly addressing one of the largest sources of organ non-utilization.

ANN-Based Liver Matching (MADR-E). The large Spanish multicenter study (Model for Allocation of Donor and Recipient in España) demonstrated that artificial neural networks considering donor, recipient, and match characteristics could predict 3-month graft survival, complementing the MELD allocation system in 55% of cases.

2.2 Individualized Treatment Effect (ITE) Approaches

The most sophisticated line of research reframes organ matching as an individualized treatment effect problem — recognizing that each organ is a unique "treatment" and each patient responds differently to each possible organ.

OrganITE (NeurIPS 2020). Developed by Berrevoets et al. at the van der Schaar Lab, OrganITE introduces an allocation methodology that considers: (1) the unique features of each patient, (2) the expected survival of each patient without receiving an organ, (3) the benefit each patient would receive from each specific available organ, and (4) the relative rarity of each patient's "optimal" organ type. By modeling organ scarcity explicitly, OrganITE demonstrated gains of up to an additional year of life expectancy across the population compared to greedy best-match approaches. OrganITE adopts a hybrid policy — sitting between pure best-match and sickest-first — by balancing individualized treatment effect, quality of match, and rarity of organ.

Matching Representations (AISTATS 2021). Xu et al. proposed learning representations that cluster donor features and apply donor-invariant transformations to predict compatibility, addressing the challenge that we only observe outcomes for matches that were actually made (factual outcomes), not for the countless possible matches that were never tried (counterfactual outcomes). The model uses representation learning to handle selection bias inherent in historical allocation data.

OrganSync. An extension that formulates organ allocation as a queueing problem, accounting for the stream nature of organ arrival — while a currently available organ might benefit a patient, a better-matched organ might arrive in the future. OrganSync uses interpretable models to remain auditable for clinical adoption.

2.3 Additional AI Applications Along the Transplant Pipeline

Beyond matching itself, AI has demonstrated value across the transplant pipeline:

- **Organ quality assessment:** Deep learning on pre-implantation kidney biopsy whole-slide images improves identification of tissue components and predicts graft function, achieving AUC improvements from 0.66 to 0.80 when biopsy images are combined with clinical features.
- **HLA matching and crossmatch prediction:** AI models integrating multi-omics data and transcriptomic profiling enhance prediction accuracy for histocompatibility and crossmatch outcomes beyond traditional serological methods.
- **Organ acceptance prediction:** XGBoost and other ensemble models predict whether a transplant center will accept an organ offer, enabling smarter parallel offer strategies.
- **Post-transplant monitoring:** ML models predict delayed graft function, acute rejection episodes, and long-term graft loss using postoperative clinical data.

- **Donor identification:** Automated detection of potential organ donors from hospital electronic health records using temporal clinical data patterns.

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3. The Core ML Challenge: Why This Is Hard

Organ matching is not a standard supervised learning problem. It has several unique challenges that must be addressed in any credible approach.

3.1 Counterfactual Outcomes

We only observe outcomes for donor-recipient pairs that were actually matched — the factual outcomes. We never observe what would have happened if a different patient had received that organ. Any ML model must address this counterfactual inference problem, typically through techniques from causal inference: propensity score weighting, balanced representations, instrumental variables, or semi-synthetic data generation.

3.2 Selection Bias

Historical matching data reflects the biases of past allocation policies. If certain donor-recipient combinations were never tried (e.g., older donors rarely matched with younger recipients), the model has no data on those combinations. Models trained naively on this data will simply learn to replicate existing policy, not improve on it.

3.3 High-Dimensional Matching Space

Each organ has dozens of features (donor age, BMI, cause of death, HLA type, cold ischemia time, organ-specific measurements). Each candidate has dozens more. The interaction space is combinatorially vast. Feature engineering and representation learning are critical to making this tractable.

3.4 Time-Dynamic Decisions

A patient's optimal strategy changes over time — declining a marginal organ today might be rational if a better organ is likely to arrive soon, but not if the patient's condition is deteriorating rapidly. The model must account for the sequential, queueing nature of organ arrival and patient deterioration.

3.5 Multiple Competing Objectives

Organ allocation serves multiple objectives that can conflict: maximizing total life-years saved, prioritizing the sickest patients, ensuring geographic and demographic fairness, minimizing organ waste, and respecting patient autonomy. No single objective function captures the full policy intent. The model must support multi-objective optimization and transparent tradeoff analysis.

3.6 Explainability and Trust

Clinicians, patients, and policymakers must be able to understand why the model recommends a particular match. Black-box neural networks, however accurate, face significant adoption barriers. The approach must incorporate explainable AI (XAI) techniques — SHAP values, attention mechanisms, counterfactual explanations, or inherently interpretable model architectures like optimal classification trees.

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4. Proposed Approach: AI-Augmented Matching for OPTN

4.1 Design Philosophy

The AI matching system operates as a **decision-support layer alongside the deterministic policy engine** — not as a replacement for policy-driven allocation. The deterministic policy engine (described in the Architecture document) enforces OPTN allocation rules as approved by the Board and the HHS Secretary. The AI layer augments the process by providing predictions, risk scores, and recommendations that inform — but do not override — the policy-driven ranking.

This is critical for regulatory, ethical, and practical reasons: allocation rules must be transparent, auditable, and subject to public governance. AI models can improve the *inputs* to those rules (better severity scores, better outcome predictions, better donor quality indices) and can identify opportunities the rules miss (organs likely to be declined that could benefit a patient further down the list), but the rules themselves remain human-authored and policy-governed.

4.2 Three-Layer Architecture

The AI matching system is structured in three layers, each with increasing sophistication and decreasing clinical autonomy:

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Layer 3: Allocation Policy Optimizer (Advisory)
   |  Simulates proposed policy changes, recommends rule modifications
   |  Fully advisory — requires OPTN Board governance approval
   |
Layer 2: Donor-Recipient Match Scoring (Decision Support)
   |  Predicts pair-level outcomes, transplant benefit scores
   |  Displayed to clinicians alongside policy-driven ranking
   |
Layer 1: Enhanced Component Scores (Integrated)
   |  ML-powered replacements for MELD, KDPI, EPTS, etc.
   |  Plugs directly into the policy engine after clinical validation
   |
   ▼
Deterministic Policy Engine (Authoritative)
   Executes OPTN allocation rules as approved by Board/HHS
```

4.3 Layer 1: Enhanced Component Scores

Goal: Replace or augment existing scoring models (MELD, KDPI, EPTS, Lung CAS) with ML models that use the same input data but produce more accurate predictions.

Approach:

- Train gradient-boosted tree ensembles (XGBoost/LightGBM) and optimal classification trees on the full SRTR historical dataset for each organ type.
- Primary prediction targets: 90-day waitlist mortality/removal (liver), 1-year and 5-year graft survival (kidney), 1-year post-transplant survival (heart, lung).
- Use all available pre-transplant clinical variables as features — not just the 3-4 variables in current models.
- Validate on held-out temporal cohorts (train on years N through N+K, validate on year N+K+1) to prevent data leakage and ensure the model generalizes to future patients.
- Deploy inherently interpretable models (optimal classification trees, as in OPOM) or ensemble models with mandatory SHAP explanations for every score.

Output: An enhanced severity/priority score for each candidate, and an enhanced donor quality index for each organ. These plug directly into the existing policy engine's allocation rules — they are drop-in replacements for MELD, KDPI, etc.

Clinical governance: Each enhanced score requires validation by the relevant OPTN organ-specific committee (liver, kidney, heart, lung), simulation against historical data using OPTN simulation models (LSAM, KPSAM, TSAM equivalents), public comment, and Board approval

before production use. The existing score runs in parallel during a shadow period, with discrepancies analyzed by the Data Advisory Committee.

AWS implementation:

- Training: SageMaker Training Jobs with hyperparameter optimization on historical SRTR data stored in S3 (Iceberg tables).
- Serving: SageMaker Real-Time Endpoints behind the Matching Service, invoked during match run scoring.
- Monitoring: SageMaker Model Monitor for data drift, prediction drift, and feature attribution drift, with automated alerts.
- Retraining: Quarterly automated retraining pipeline (SageMaker Pipelines) using the latest SRTR data, with human-in-the-loop validation gate before production promotion.

4.4 Layer 2: Donor-Recipient Match Scoring

Goal: Predict the expected outcome for a specific donor-recipient pair — moving beyond patient-only severity scores to pair-level compatibility predictions.

Approach:

- Train a **Transplant Benefit Score (TBS)** model that estimates: (a) expected post-transplant survival at 1 year and 5 years for this specific donor-recipient pair, and (b) expected survival without transplant (remaining on the waitlist) for this specific recipient. The TBS is defined as the difference: TBS = E[survival with this transplant] - E[survival without transplant].
- Model architecture: A **representation learning approach** (following the Matching Representations paradigm) that learns joint donor-recipient embeddings. The donor feature space and recipient feature space are each mapped to a shared latent space through separate encoder networks, and a compatibility function predicts outcomes from the joint representation.
- Address counterfactual outcomes using balanced representation learning (penalizing distributional distance between treated and untreated representations) and semi-synthetic validation.
- Address selection bias in historical data through inverse propensity weighting using the known historical allocation policy as the propensity model.
- For each prediction, generate SHAP-based explanations identifying which donor features, which recipient features, and which interaction features most influenced the predicted benefit.

Output: For each candidate on the match run, the system displays: (1) the policy-driven rank from the deterministic engine, (2) the predicted Transplant Benefit Score, (3) the predicted 1-year and 5-year graft survival probability, (4) a confidence interval, and (5) the top contributing factors. This information is presented to the transplant center surgeon reviewing the organ offer — they see both the policy rank and the AI-predicted benefit.

Critically, the TBS does not change the rank order. The deterministic policy engine determines the rank. The TBS provides additional clinical context that may inform a center's accept/decline decision, or an OPO coordinator's strategy for parallel offers on hard-to-place organs.

Organ acceptance prediction (sub-model): A separate model predicts the probability that each center on the match run will accept the offer, based on historical center-specific acceptance patterns, donor characteristics, time of day, and ischemia time estimates. This enables "smart offers" — when the model predicts high decline probability for the top-ranked centers, the Offer Service can proactively initiate parallel offers to lower-ranked but likely-to-accept centers, reducing cold ischemia time and organ waste.

AWS implementation:

- Training: SageMaker Training Jobs with distributed training for the deep representation learning model. Semi-synthetic counterfactual data generation via SageMaker Processing Jobs.
- Serving: SageMaker Real-Time Endpoints (GPU instances for the deep model) or Bedrock custom model endpoints, invoked during the offer presentation workflow.
- Embedding store: Feature vectors stored in Amazon OpenSearch (k-NN) for fast similarity search when rapid organ-patient compatibility lookup is needed.
- Explainability: SHAP values computed in real-time via a SageMaker Processing endpoint and rendered in the Offer Service UI.

4.5 Layer 3: Allocation Policy Optimizer

Goal: Provide OPTN policymakers with simulation-based tools to evaluate the impact of proposed allocation policy changes before implementation.

Approach:

- Build a **digital twin of the OPTN** — a simulation environment that replays historical donor arrivals, candidate additions/removals, and match runs under any specified allocation policy.
- The simulator uses the Layer 1 and Layer 2 ML models as its outcome prediction engine, replacing the statistical models in legacy simulation tools (LSAM, KPSAM).

- Policy analysts define proposed rule changes (e.g., "change the kidney allocation distance threshold from 250nm to 500nm," "replace MELD with the enhanced ML severity score," "add a longevity matching component for kidneys with KDPI < 20%").
- The simulator runs the proposed policy against 5+ years of historical data and reports projected outcomes: transplant volumes, waitlist mortality, graft survival rates, organ utilization rates, geographic distribution effects, and demographic impact analysis.
- A **LLM-powered policy copilot** (via Bedrock) assists analysts in translating natural-language policy intent into the simulation configuration, interpreting results, and generating summary reports for the Board.

Output: A policy simulation report showing the projected impact of the proposed change compared to the current policy, with confidence intervals and sensitivity analysis. This report goes to the OPTN Board, relevant committees, and public comment — the same governance process as today, but with dramatically better predictive analytics backing the analysis.

AWS implementation:

- Simulation engine: AWS Batch (compute-optimized instances) running the digital twin simulation in parallel across multiple historical cohorts.
- Policy configuration: DynamoDB-backed policy configuration store with versioning.
- LLM copilot: Bedrock foundation model with Bedrock Knowledge Bases containing OPTN policy documents, historical simulation results, and the SRTR data dictionary.
- Reporting: SageMaker Processing Jobs generating visualization outputs, rendered in QuickSight or a custom dashboard.

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5. Data Requirements

5.1 Training Data Sources

Data Source	Contents	Use
OPTN/SRTR historical dataset (1987–present)	All donors, candidates, transplants, outcomes, follow-up	Primary training and validation
CMS Medicare claims	Post-transplant healthcare utilization, death records	Outcome enrichment, long-term survival validation
NTIS Death Master File	Death events for waitlist candidates who died without transplant	Waitlist mortality outcomes

Data Source	Contents	Use
OPTN match run archives	Full offer chains — every offer, acceptance, decline, reason code	Acceptance prediction model training
Histocompatibility lab data	HLA typing, crossmatch results, PRA/CPRA panels	Immunologic compatibility features
Donor imaging (where digitized)	CT, echocardiogram, biopsy slides	Organ quality assessment models (future)

5.2 Feature Engineering

Donor features (~60 variables): Age, sex, BMI, cause of death, mechanism of death, donor type (DBD/DCD/living), blood type, HLA typing, serology panel, organ-specific measurements (creatinine, eGFR, bilirubin, ejection fraction, PaO2, etc.), medical/social history, recovery surgeon assessment, cold ischemia time estimate, pump parameters (if machine perfusion used).

Recipient features (~80 variables): Age, sex, BMI, blood type, HLA typing, CPRA, diagnosis, comorbidities (diabetes, hypertension, cardiovascular disease), prior transplants, dialysis duration and type, functional status, lab panel (albumin, creatinine, bilirubin, INR, sodium), listing date, time on waitlist, listing center, insurance type, geographic location.

Pair-level features (~30 derived variables): HLA mismatch count (A, B, DR, DQ loci), ABO compatibility type, age difference, BMI difference, size match (weight ratio), donor-recipient distance, estimated cold ischemia time, crossmatch result, donor-specific antibody status.

Temporal features: Time on waitlist, rate of clinical deterioration (slope of lab values over recent measurements), season, time of day (correlates with center staffing and decision patterns).

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6. Model Governance and Safety Framework

6.1 Validation Requirements

Every ML model deployed in the OPTN matching pipeline must pass the following validation gates:

1. **Technical validation:** Out-of-sample AUC, calibration (Brier score), and discrimination metrics on temporal hold-out cohorts. Performance must exceed the current scoring system across all organ types and major patient subgroups.
2. **Simulation validation:** Allocation simulation using the Layer 3 digital twin, demonstrating improvement (or non-inferiority) in transplant volume, waitlist mortality, and graft survival compared to the current policy.

3. **Clinical validation:** Review by the relevant OPTN organ-specific committee and the Data Advisory Committee. Clinical experts assess whether the model's feature importances and decision boundaries align with clinical knowledge.
4. **Bias audit:** Systematic analysis of model performance across demographic subgroups (age, sex, race/ethnicity, geography, insurance type, diagnosis). Any statistically significant performance disparity must be documented, explained, and mitigated before deployment.
5. **Shadow deployment:** Minimum 6-month parallel run where the ML model generates scores alongside the existing scoring system, with automated discrepancy reporting and clinical review of divergent cases.
6. **Public comment:** Per OPTN Final Rule requirements, any change to allocation scoring that would affect patient ranking must go through the standard OPTN public comment process.
7. **Board approval:** The OPTN Board of Directors must approve the model for production use, with documented acceptance criteria and a rollback plan.

6.2 Continuous Monitoring

Post-deployment, every model is subject to:

- **Data drift detection:** SageMaker Model Monitor tracks statistical distribution shifts in input features. If drift exceeds thresholds, alerts are generated and the model is flagged for retraining evaluation.
- **Prediction drift detection:** Monitoring of output score distributions over time. Sudden shifts may indicate upstream data quality issues or clinical practice changes.
- **Outcome monitoring:** Quarterly comparison of predicted vs. actual outcomes (graft survival, waitlist mortality) to validate ongoing calibration.
- **Fairness monitoring:** Ongoing tracking of model performance across demographic subgroups, with automated alerts for emerging disparities.
- **Incident reporting:** Any case where a clinician identifies a model recommendation that appears clinically inappropriate is logged, investigated, and fed back into model improvement.

6.3 Explainability Requirements

All models must provide:

- **Global explanations:** Feature importance rankings and partial dependence plots published in the model documentation.
- **Local explanations:** For every individual score or prediction, SHAP values or equivalent attributions identifying which features drove the prediction and by how much.

- **Counterfactual explanations:** For Layer 2 pair-level predictions, the ability to show "what would the predicted outcome be if this donor feature were different?" — supporting clinical reasoning about marginal organs.
- **Uncertainty quantification:** Every prediction must include a confidence interval or prediction interval, communicated to the clinician in a format validated through user research.

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7. Ethical and Regulatory Considerations

7.1 OPTN Final Rule Compliance

The OPTN Final Rule (42 CFR Part 121) requires that organ allocation policies be based on "sound medical judgment" and that "policies shall be designed to avoid wasting organs, to avoid futile transplants, to promote patient access to transplantation, and to promote the efficient management of organ placement." ML models must demonstrate alignment with these statutory objectives. Any model that changes patient ranking in the allocation sequence constitutes a policy change subject to the full OPTN governance process.

7.2 Transparency Obligation

The OPTN operates under public trust. All model architectures, training data descriptions, validation results, and performance metrics must be publicly disclosed. Model source code should be available for independent audit. The model must not rely on proprietary algorithms that cannot be inspected by regulators, researchers, or patient advocates.

7.3 Human Override

Clinicians must always retain the ability to override model recommendations. The system must make clear that AI-generated scores and predictions are advisory, not directives. Transplant surgeons make the final accept/decline decision for each organ offer, and that authority is preserved.

7.4 Historical Bias in Training Data

OPTN historical data reflects decades of allocation decisions made under policies that have been criticized for various biases. Models trained on this data will inherit those biases unless explicitly corrected. The approach must include: (a) causal inference techniques to separate clinical signal from policy-induced bias, (b) sensitivity analysis showing model performance under alternative historical policies, and (c) prospective monitoring for perpetuation of historical patterns that conflict with current policy intent.

7.5 Avoiding Automation Bias

There is a well-documented risk that clinicians will defer to AI predictions even when their own clinical judgment suggests otherwise ("automation bias"). The system must be designed to present AI information as one input among many, use calibrated language (probabilities with uncertainty, not definitive statements), and support — not supplant — clinical reasoning.

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8. Key Research References

Model/Study	Organ	Key Finding
OPOM (Bertsimas et al., 2019)	Liver	ML classification trees reduced simulated annual waitlist mortality by ~418 deaths vs. MELD
UK-DTOP (2024)	Kidney	AI tool achieved AUCROC 0.74 vs. 0.62 for UK-KDRI; identified 5th donor risk cluster
OrganITE (Berrevoets et al., NeurIPS 2020)	Multi-organ	ITE-based allocation accounting for organ scarcity gained up to 1 additional life-year per patient
Matching Representations (Xu et al., AISTATS 2021)	Liver	Representation learning for counterfactual match outcome estimation
OrganSync (van der Schaar Lab, ICML 2021)	Multi-organ	Interpretable queueing-aware allocation policy
Trees of Predictors (Yoon et al., 2018)	Heart	Personalized survival prediction improving on clinical risk scores
MADR-E (Briceño et al., 2014)	Liver	ANN for D-R matching complemented MELD in 55% of cases
Stanford DCD Model (2025)	Liver	ML prediction of donor death timing reduced futile procurements by ~60%
Kidney Graft ML Meta-Analysis (PMC, 2023)	Kidney	SVM and RF models outperform Cox regression; hybrid models perform best
Deep Learning Biopsy (Frontiers, 2021)	Kidney	Adding biopsy WSI images to clinical features improved prediction AUC from 0.66 to 0.80

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9. Summary

AI-powered organ matching represents the single highest-impact application of machine learning in the modernized OPTN. The research base is mature, with multiple validated models demonstrating meaningful improvements over current scoring systems — improvements that translate to hundreds of saved lives per year in simulation.

The approach proposed here is deliberately conservative in its governance model. AI does not make allocation decisions. AI improves the scoring inputs to the policy engine (Layer 1), provides additional clinical context to inform accept/decline decisions (Layer 2), and enables better policy design through simulation (Layer 3). Every layer preserves clinician authority, requires rigorous validation, and follows the OPTN's established governance process.

The key technical innovations are: (1) moving from patient-only severity scores to donor-recipient pair-level benefit predictions, (2) addressing counterfactual outcomes and selection bias in historical data through causal inference techniques, (3) incorporating organ scarcity and the sequential nature of organ arrival into the optimization, and (4) maintaining full explainability through SHAP attributions, interpretable model architectures, and uncertainty quantification.

If implemented with the phasing and governance framework described above, this approach can reduce waitlist mortality, increase organ utilization, decrease cold ischemia times through smarter offers, and provide OPTN policymakers with unprecedented analytical capability to design and evaluate allocation policies — while maintaining the transparency and public accountability that the transplant system demands.